

SPPH 604 — In-class lecture slide plans (Weeks 1–7)

Slide-by-slide **blueprints** (not finished decks) for the weekly class session. Weeks 2–7 run **Tuesday 9:30 am–12 pm (150 minutes)**, ahead of that week’s Thursday lab (10 am–12 pm), so the lecture and lab feel continuous. **Week 1 is the exception:** there is no Tuesday session and no lab — the course opens with a **two-hour Thursday session** (Sept 10, 10 am–12 pm, by Zoom). Each lecture teaches the concept and theoretical issue of the demonstration paper (**MASLD / NHANES**, Kueh et al., BMJ Open 2026) *before* students go into that week’s lab.

Every plan has the same shape:

- **Learning objectives** (3–5)
- **Continuity** — recap of last week’s lab + what today sets up
- a numbered **slide blueprint** — each slide has content bullets, a “**MASLD hook**” tying it to the paper, and an approximate minute count (with a break marker)
- a closing “**Bridge to today’s lab**” slide stating exactly what students do in Lx and how it becomes their increment Px

The seven weeks

Week	Lecture	Feeds lab	Feeds increment
1	Course intro + procedures (+ 1 paper teaser)	— (setup)	begins M0 shortlisting
2	NHANES data sources; reading Methods critically	L1 Data wrangling	M0 shortlist
3	Cohort construction — FLI, MASLD criteria, eligibility	L2 Analytic data	P1 eligibility (M0 locked)
4	Confounding & DAGs; Table 1; variable roles	L3 Confounding	P2 Table 1
5	Interaction & effect modification	L4 Interaction	P3 effect modification
6	Complex survey design (weights/strata/PSU)	L5 Survey analysis	P4 design-aware estimate
7	Missing data (MCAR/MAR/MNAR; MI)	L6 Missing data	P5 missing-data sensitivity

Design notes

- **Continuity is the point.** Slide 1 of each week recaps where the project stands; the last slide hands off directly into the lab and the formative increment.
- **The paper is the running example.** Concepts are taught through the MASLD reproduction: the typed cohort funnel, FLI circularity, how far adjustment moves a hazard ratio and what that move can and cannot mean, the sex interaction that is null unweighted and moves under the design, and what survey weighting does to an estimate (it can move it in either direction) and to its interval. The cohort size is the first lesson rather than a verified match — the paper reports $N \approx 6,300$, our full analytic file is 6,371, and accounting for the difference is the skill being taught.
- **Two analyses of one cohort.** The **paper replication** (Weeks 2–3, P1–P5, M0–M4) keeps the paper’s four phenotype groups and its Model 1 / Model 2 covariates. The **lecture’s exposure analysis**

(Weeks 4–7) asks about obesity, and about high waist fat within obesity strata, adjusted for the pre-exposure set L (age, sex, race/ethnicity, current smoking, sedentary time), mostly on 5,911 rows. The two can give different numbers for the same contrast, mainly because the adjustment set differs, so every MASLD number is labelled with its comparison, rows, adjustment set and weighting.

- **The weeks build toward M1.** The demonstration paper runs through Weeks 3–5 as sustained practice in replication and methodological interrogation, so the analytic judgment **M1** asks each group to make rests on skills they have already rehearsed — the judgment itself stays theirs.
- **Timing convention — budget to the clock.** The session length is the cap: **150 min** for a Tuesday class (Weeks 2–7), **120 min** for the Week 1 Thursday opener. A plan’s own minute markers are the whole budget and the only accounting: they include the break, and nothing outside them is counted, because the untimed **Learning objectives** and **Continuity** blocks at the head of every plan are covered inside that week’s Slide 1. Markers may sum *to* the cap; they may never sum *past* it. Whatever is left over is **float** — minutes the room spends on questions and the vote-and-argue moments. Buffer is carried *inside* the plan, never left as unallocated time at the end: each plan names one recap or preview slide as its **designated cut**, so an overrun costs a slide chosen in advance instead of the bridge into the lab. The plan file is the budget of record; the decks’ speaker-note timings re-allocate a plan’s minutes and are not summed here.

Checking the timings

Each plan states its own total on its **Duration:** line. That line is the claim; this is the check, and it re-derives all seven weeks at once whenever a plan is retimed (Git Bash, from the repo root):

```
python -c "import re,glob;P=re.compile(r'(?m)^(?:\s*\s.*?(~?(\d+) ?min\)|\s*[-*]\s*(~?(\d+) ?min\))?)"
```

Read each total against that week’s cap — 120 for Week 1, 150 for Weeks 2–7. Only what the command matches counts as a marker: minutes written into a plan’s prose are not part of the budget, so a total counted by eye can come out higher than the real one. A week over its cap owes a trim in its own file, not a footnote here. This README publishes no table of the current totals on purpose — it would be a frozen snapshot of seven files it does not own, and stale the first time one of them is retimed.

Reading the demonstration paper (bmj_open_revealjs/)

A standalone reveal.js deck that decomposes the paper’s published aim into a PECOT question, states how MASLD was actually ascertained (the Fatty Liver Index, not imaging), and lists every NHANES variable the study uses beside the component file it comes from. Intended to be worked through **before L1**.

Open it at ehsanx.github.io/SPPH604-2026W/lectures/bmj_open_revealjs/.

Built decks (slides/)

The plans have been turned into finished **Quarto revealjs decks** in `slides/` — one `.qmd` per week plus a self-contained rendered `.html` (open in any browser; arrow keys to navigate, **S** for speaker notes, **F** for fullscreen). Each deck has a title slide, objectives, a continuity slide, the plan’s content slides with **MASLD** running-example callouts, and detailed **speaker notes** (timings + teaching prompts) under each slide. The Kaplan–Meier figure is embedded on the weeks that discuss the survival result (2, 5, 6). Each `.html` is the rendered form of the `.qmd` beside it, so present from the `.html` and treat the `.qmd` as the source. If you adapt a deck, change the `.qmd` and re-render it with Quarto rather than editing the `.html`. From this directory:

```
quarto render slides/Week5_interaction_slides.qmd
```

`img/` holds the embedded figures. The two reproduction figures (`figure1_km.png`, `figure2_rcs.png`) are copied from `reproduction/output/figures/`; the w4–w7 lecture figures accompany the lecture analyses.